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QLoRA Fine-Tuning for Consumer Complaint Classification

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Abstract: Consumer financial complaint classification requires balancing adaptation gains against lighter supervised alternatives. We compare Qwen3-8B and Llama 3.1-8B-Instruct before and after explicit 4-bit QLoRA adaptation on the same fixed 10,000 training and 2,000 validation examples. Data allocation uses model-visible input-identity checks and excludes conflicting training/validation records from fitting while retaining the evaluation records. On the 15,428-row internal test, mean macro-F1 across three optimisation seeds rises from 0.6295 to 0.8331 for Qwen and from 0.6247 to 0.8402 for Llama. Paired group-bootstrap gains are 0.2035 [0.1934, 0.2141] and 0.2155 [0.2054, 0.2257]. Validation-tuned TF-IDF/Logistic Regression reaches 0.8068 with 10,000 examples and 0.8476 with the larger 120,502-row training pool. Increasing available narrative text from 500 to 1,500 characters improves adapted-model macro-F1 by 0.0072 and 0.0055, but does not consistently improve zero-shot performance. Controlled batch-one measurements show additional mean end-to-end time of 0.1224 and 0.0802 seconds for adaptation. These retrospective, source-related evaluations support a budget-conditional adaptation trade-off, not universal classification superiority or deployment acceptability.

Key words: Consumer complaint classification, large language models, QLoRA, parameter-efficient fine-tuning, model-visible overlap, paired group bootstrap, CFPB

1. Introduction

The Consumer Financial Protection Bureau (CFPB) complaint database is a large public corpus of consumer narratives describing financial disputes, product failures, debt collection issues, credit reporting problems, and banking complaints. Accurate classification of such narratives is important for regulatory monitoring, institutional triage, and the prioritisation of consumer-service workflows. The task is challenging because complaint narratives often mention multiple financial products, contain noisy informal descriptions, and must nevertheless be mapped to a single constrained product category.

Recent work has explored large language models (LLMs) for zero-shot complaint classification. In zero-shot settings, models receive the candidate categories in the prompt and must return a valid label without task-specific training. This is attractive for complaint triage because new product taxonomies or institutional routing schemes may be introduced before large labelled datasets are available. However, zero-shot classification also raises practical issues: models can confuse overlapping financial products, output labels outside the allowed taxonomy, and show sensitivity to prompt and deployment details. Once a local zero-shot baseline is established for a given open-weight model, the next methodological question is how much performance can be gained through parameter-efficient fine-tuning under a fixed labelled-data budget.

This paper addresses that question. The central novelty is not a new model architecture or a new fine-tuning algorithm. Güllü et al. already fine-tuned GPT-2 and Llama 3.1-8B for complaints, including the same five financial categories [14]; Roumeliotis et al. studied zero-shot financial complaint classification [15]. Our contribution is a reproducible applied evaluation of the paired zero-shot-to-QLoRA transition, including validation-tuned lighter classifiers and the benefits and costs of adaptation. The primary comparison uses fixed

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data selections and matched inference settings. Supplementary Section S8 details configuration screening and auxiliary dependence analyses.

The contributions are as follows:

1. We compare base zero-shot and QLoRA-adapted models within the same Hugging Face quantised inference pipeline, holding prompt, labels and main input conventions fixed within each model family.
2. We audit model-visible inputs, quarantine conflicting training/validation records, and preserve evaluation records and the source-related benchmark views with their dependence explicitly accounted for.
3. We evaluate Qwen3-8B and Llama 3.1-8B across optimisation seeds 42, 43 and 44 on fixed 10,000-example training and 2,000-example validation selections, alongside tuned sparse and encoder baselines.
4. We report internal and benchmark performance, paired group-bootstrap intervals, separate seed variation, off-label behaviour and class-level error patterns.
5. We measure same-checkpoint longer-input effects and controlled fresh-call latency to quantify conditional adaptation trade-offs.

2. Related work

Classical consumer complaint classification has relied on sparse lexical features, support vector machines, logistic regression, and later transformer encoders. BERT-style models enabled strong supervised text classification by adapting contextual representations to task-specific labels [1]; BERT, DistilBERT, and RoBERTa remain important supervised baselines for complaint categorisation [1–4]. Complaint analytics has also been studied across food-industry categorisation, telecom complaint prediction, multicriteria complaint prioritisation, CFPB topic modelling, and financial-complaint interpretability, sentiment, cause, emotion, and severity analysis [5–9]. These studies show that supervised learning can be effective for complaint taxonomies, but they do not directly address how modern open-weight generative models should be adapted and evaluated for the same task.

Recent work has made LLMs relevant to both complaint generation and complaint processing. Large-scale studies report that consumers increasingly use LLMs to draft clearer financial complaints [10, 11], suggesting that complaint corpora are themselves being reshaped by generative AI. LLM-based frameworks have also been investigated for complaint categorisation and customer-service benchmarking [12, 13]. Two studies are particularly close to the present work and require separate comparisons.

Güllü et al. [14] compare CNN, LSTM, BERT, RoBERTa, GPT-2, and Llama 3.1-8B on two datasets: 5,629 Comcast complaints with rating labels and a second dataset reported to contain 162,421 financial complaints in five product categories. The latter categories correspond to the financial taxonomy studied here. Thus, their contribution is not restricted to telecommunications, and the present work cannot be distinguished merely by applying Llama to financial complaints. Their method uses an 80/20 train/test division; Table V of their paper specifies three training epochs, a 512-token maximum, AdamW, cross-entropy loss, and weighted metric averaging for the LLMs. A LoRA or 4-bit quantisation configuration, a paired zero-shot baseline for each adapted checkpoint, and a three-seed uncertainty summary are not described in that report. Absence of these details is treated as a reporting difference, not evidence that an undocumented procedure was not performed.

Our distinction from Güllü et al. is the adaptation question and the explicit protocol: 4-bit NF4 LoRA adaptation with completion-only loss for JSON label generation, a fixed 10,000-example training budget, two 8B model families evaluated before and after adaptation in the same inference pipeline, and repeated fine-tuning seeds. Table 1 summarises these differences. Their published financial-dataset F1 scores should not be ranked against our macro-F1 scores: metric averaging, training budgets, and test construction differ. Nor does their

published row count establish exact record-level identity with our 162,415-row source file.

Table 1. Protocol comparison with Güllü et al. [14]. This is a comparison of reported designs, not a cross-paper performance ranking. "Not described" refers only to the information available in that paper.

Aspect	Güllü et al.	Present study
Task and data	Comcast rating classification and five-category financial complaints; financial dataset reported as 162,421 records.	Five-category financial complaints from a 162,415-row source file; no new corpus or taxonomy is claimed.
Comparison	Supervised comparison of six architectures, including GPT-2 and Llama 3.1-8B.	Paired zero-shot versus QLoRA adaptation for Qwen3-8B and Llama 3.1-8B; TF-IDF and encoders as supervised comparators.
Adaptation	LLM training with cross-entropy loss; LoRA and 4-bit settings not described.	Explicit 4-bit NF4 LoRA configuration and completion-only loss for generated JSON labels.
Data allocation	80/20 train/test division; separate validation selection and text-group overlap audits not described.	Model-visible conflicts are quarantined; fixed training/validation selections and evaluation records are retained across all model runs.
Budgets	Three LLM epochs and a maximum of 512 tokens.	Two epochs on 10,000 selected training rows per seed; 500 narrative characters within a 512-token prompt budget.
Reporting	Accuracy, precision, recall, and weighted F1; three-seed uncertainty summaries not described.	Accuracy, macro-F1, weighted F1, and results across three fine-tuning seeds; separate analysis of adaptation gains.

Roumeliotis et al. [15] address a different question: zero-shot classification with 14 standard and reasoning LLMs on a balanced 1,000-record CFPB-derived evaluation set. Their preprocessing removes narratives exceeding 500 characters, whereas the present protocol truncates narratives to a maximum of 500 characters and also evaluates the preserved 250/500-character benchmark variants. Their study motivates the five-label JSON-prompt setting, but does not evaluate the paired change caused by task-specific QLoRA adaptation. Our additional model-family and seed comparisons therefore extend the evaluation question rather than claim the first use of this taxonomy or a directly comparable improvement over their published scores.

PEFT provides a middle path between zero-shot inference and full fine-tuning. LoRA learns low-rank adapter matrices while freezing pretrained weights [16], and QLoRA enables quantised LLM adaptation through 4-bit NormalFloat quantisation, double quantisation, and LoRA adapters [17]. Other PEFT methods such as prefix tuning and $(IA)^3$ are also viable [18, 19], but QLoRA was selected here for 4-bit adaptation of 8B-class instruction models on 16 GB GPUs. The empirical question is how much same-model adaptation changes classification under the stated budget and how those gains compare with lighter supervised alternatives. It also requires distinguishing source-text grouping from model-visible separation and accounting for overlap among benchmark subsets. These are evaluation requirements for the present study, not claims of a new LoRA or QLoRA algorithm or of the absence of related fine-tuning work.

3. Materials and methods

3.1. Data provenance and preserved evaluation views

The processed CFPB-related resource is attributed to Tiwari on Kaggle [20, 26]. Its local archive contains 162,421 rows. The working file is identical by index, label and narrative for the first 162,415 rows; six trailing records contain only the placeholder narrative `name` and are absent. The archive-to-working-file correspondence identifies the analysed source records; the available records do not establish the undocumented preprocessing

decisions. Removing ten rows with missing fields leaves 162,405 usable records. Excluding 8,123 benchmark-related source rows leaves a partitioning pool of 154,282 records. The upstream CFPB-to-Kaggle extraction and product-aggregation code is unavailable; source hashes identify the exact analysed input.

The five inherited labels are retail banking, credit reporting, credit card, mortgages and loans, and debt collection. No new upstream product aggregation is performed. Twelve preserved benchmark files contain 1,000 appearances each: sampling-seed identifiers 1/2/3, balanced/imbalanced sampling, and 250/500-character views. Balanced files contain 200 examples per class. Each imbalanced file contains 511 credit-reporting, 182 debt-collection, 107 mortgage/loan, 100 card and 100 banking examples. These files are byte-identical to their source-archive members. The sampling implementation is unavailable; files and construction manifests are supplied. They are source-related benchmark views, not independently collected external data.

3.2. Model-visible separation and fixed selections

Full-narrative equality alone is insufficient for a classifier that sees truncated text: distinct narratives can share the same visible prefix or token sequence. Data allocation therefore checks the representations supplied to the models rather than relying solely on full-text duplicate detection. Candidate roles use an 80/10/10 allocation with seed 42 and full-normalised-narrative grouping; the input-identity constraints below determine which fitting records are retained.

Label-blind connected components link full-normalised narratives, normalised first-500-character inputs and reconstructed 512-token inputs of the tokenizers used to construct the allocation. A conservative normalised 250-character prefix guard links benchmark-related inputs. Priority is given to the internal and benchmark evaluation records. Conflicting training and validation rows are quarantined; remaining validation components take priority over conflicting training rows. Evaluation records are not moved into fitting sets. Table 2 gives the counts. Prefix components are conservative text dependencies, not verified complaint identities.

Table 2. Data allocation with model-visible overlap screening. Evaluation records are excluded from fitting.

Partition	Candidate rows	Excluded	Retained	Groups
train	123427	2925	120502	93609
validation	15427	307	15120	11925
internal test	15428	0	15428	12099

Twenty-five retained-partition checks find zero shared audited representations. The five actual pinned tokenizers are then independently audited on 39,428 inputs each: 10,000 train, 2,000 validation, 15,428 internal and 12,000 benchmark appearances. Tested train/validation-to-evaluation token-identity intersections are empty. Long-input keys are also checked, and equal token inputs remain within the frozen dependency groups. This is an exact representation audit, not a guarantee of semantic or paraphrase independence.

One naturally sampled 10,000-row train selection and one 2,000-row validation selection, with selection seed 42, are fixed for every model and optimisation seed. Their IDs, order and hashes are retained. The 10k/2k budget isolates adaptation and model-family comparisons under a common labelled-data allocation; it is not an empirically selected sample-efficiency optimum. Optimisation seeds 42/43/44 change training randomness, not membership. Full-data TF-IDF uses all 120,502 retained training rows. The internal test contains 15,428 rows in 12,099 groups. Evaluation uses a fixed, inspected CFPB-related cohort and is retrospective rather than prospective. Data details and the overlap matrices are in Supplementary Sections S1 and S4.

3.3. Pinned LLMs, prompting and QLoRA

We use Qwen/Qwen3-8B [21] and meta-llama/Llama-3.1-8B-Instruct [22]. Exact 40-character commits, tokenizer/config files and weight hashes are recorded; Supplementary Section S2 gives the identities. The recorded identities apply to the models in the primary comparison.

Base and adapted models share the same five-label prompt, 500 narrative characters, a 512-token input cap, NF4 4-bit double quantisation and FP16 compute. The 500-character limit holds the paired input convention fixed; 512 is the tokenizer ceiling, not another character limit. In the usable source, 63.98% of narratives fit within 500 characters. This coverage does not establish optimality, information preservation or necessity for this GPU; the longer-input experiment measures the restriction's effects (Sections 3.6 and 4.3). The chat template is rendered once and tokenised with `add_special_tokens=False`; Qwen thinking is disabled with `enable_thinking=False`. SFT records use the identical prompt prefix and a completion mask over the target JSON/EOS. Boundary and full-target length checks reject mismatch or clipping. Greedy generation uses at most 32 new tokens, one beam and no sampling. A regular expression extracts the final `product` field; absent or invalid labels remain off-label errors. This is JSON prompting with strict parsing, not token-level constrained decoding. No retry, synonym mapping or post-hoc label repair is used.

QLoRA uses rank 16, alpha 32, dropout 0.05 and all-linear targets, with frozen base weights and completion-only cross-entropy. Each fixed-data run has two epochs, learning rate 2×10^{-4}, physical batch 1, accumulation 16, paged AdamW 8-bit, cosine scheduling, warmup 0.03 and non-reentrant gradient checkpointing. Validation loss is measured every 100 optimiser steps; the best saved validation-loss checkpoint is retained. These settings are fixed for both LLM families, without test-based configuration selection. Each model has three independently initialised optimisation refits.

The shared configuration is supported by an exploratory Qwen validation comparison of sampling strategy, learning rate, rank and epoch count. Natural sampling, learning rate 2×10^{-4}, rank 16 and two epochs have the highest recorded validation macro-F1, 0.8235, among the six screened conditions. Supplementary Section S8 specifies the screening allocation and its limitations. These configuration-screening fits are auxiliary evidence; their validation scores are excluded from the primary test-set comparison. Llama uses the same settings without family-specific optimisation.

The primary model comparison uses an RTX 5060 Ti 16 GB, driver 595.99.02, Python 3.12.14, PyTorch 2.7.1+cu128 (CUDA 12.8) [23], Transformers 4.57.6 [24], PEFT 0.17.1 [25], TRL 0.23.1, bitsandbytes 0.46.1 and Accelerate 1.10.1. Configuration-screening and auxiliary dependence analyses use the separately documented TRUBA/V100 environment; their scores are excluded from the primary comparison. Each run retains code, versions, manifests and fitted-weight hashes.

3.4. Validation-tuned supervised baselines

TF-IDF plus Logistic Regression has separate 48-candidate validation searches at the 10k and full-data budgets. Like the neural systems, both sparse budgets use the first 500 narrative characters; the full-data condition increases training rows, not visible text per row. The grid crosses unigram/bigram features, maximum features 50,000/200,000, minimum document frequency 2/5, $C = 0.25/1/4$, and no/balanced class weights. Vocabulary/IDF fit training text only. Lowercasing, Unicode accent stripping, sublinear TF and L2 normalisation are fixed. The lbfgs solver uses tolerance 10^{-5} and a 2,000-iteration cap; convergence warnings fail the run. Both winners use bigrams, minimum document frequency 5, $C = 4$ and no class weighting; feature caps are 50,000 and 200,000. Validation macro-F1 is 0.7989 and 0.8363. Three identical seed-labelled refits at each budget are

determinism checks, not stochastic optimisation replicates.

BERT-base-uncased, DistilBERT-base-uncased and RoBERTa-base use all 10k/2k selected rows. Each has six validation candidates: learning rate $2/3/5 \times 10^{-5}$ crossed with two/three epochs, search seed 42. The maximum final-epoch validation macro-F1 wins; ties within 10^{-12} use the smallest predefined candidate ID. Winners are BERT 2×10^{-5}/three epochs, DistilBERT and RoBERTa 5×10^{-5}/three epochs. Each is freshly refit with seeds 42/43/44 and evaluated at its final epoch. Main encoder inputs are 500 characters with a **512-token cap**, physical batch 4, accumulation 4, evaluation batch 16, AdamW, weight decay 0.01, linear scheduling, warmup 0.06 and FP16 training/inference. Reloaded predictions must exactly match all 2,000 saved validation predictions before test inference.

A reference-parameter sensitivity uses the encoder token caps and physical batch reported in Tables III–IV of Güllü et al., with the same selected learning rates and three epochs. Parameter provenance and all six extra refits are in Supplementary Section S7. The primary encoder baselines use 512-token inputs. Our data, 500-character text convention, PyTorch implementation and execution defaults differ from that source, so this is not an exact replication. No baseline is replaced on test performance.

3.5. Metrics, dependence and uncertainty

The natural CFPB distribution is class-imbalanced, so macro-F1 is emphasised alongside accuracy and weighted F1. Accuracy, macro-F1 and weighted F1 use the same confusion-count estimator. Off-label outputs contribute errors rather than being dropped. Points are observed-data estimates; an adapted system is the mean of three separately scored refits, not an ensemble or a pooled confusion matrix. Run variation is the sample SD (ddof $= 1$). The main and supplementary analyses report stochastic seed SD separately from conditional sampling intervals.

We use 10,000 paired percentile group-bootstrap draws (RNG seed 20261020), sampling whole groups uniformly with replacement. All 19 main prediction variants share the same group multiplicities. Internal metrics retain row weights. Benchmark analysis gives total weight one to each unique (true label, normalised visible narrative) across appearances: 5,909 units in 3,919 conservative full-text/prefix groups. Linked 250/500-character views are sampled together. The twelve overlapping file scores are dependent observations; statistical inference therefore operates on their linked groups. Paired differences are computed within each common resample. Intervals condition on fitted checkpoints and fixed selected configurations; they exclude training/model-selection uncertainty. They are descriptive 95% intervals without a family-wise multiple-comparison adjustment.

3.6. Longer inputs and controlled latency

Each base and adapted checkpoint is evaluated on the same internal IDs under 500 characters/512 tokens, 500/2,048 (token-cap control), and 1,500/2,048. No model is retrained on longer text. The full-narrative > 500 cohort is predefined (5,770 rows, 5,131 groups); cohort contrasts use paired groups. Token/character retention is recorded per input. Actual maxima are 207/403 Qwen and 224/421 Llama tokens for short/long settings, with no token-cap clipping. Deterministic generation responses may be memoised only for identical token IDs/masks with the same checkpoint, generation settings, versions and hardware. Every evaluation row remains in the analysis. Identical control inputs reuse responses; they are not independent generation replications.

Latency bypasses memoisation and measures fresh calls on 100 fixed IDs (20 per class), ten validation warmups and three repetitions per variant/condition. Input order is hash-defined per repetition. CUDA is syn-

chronised around generation; end-to-end time additionally includes tokenisation, transfer, decoding and parsing, excluding loading, warmup and disk I/O. All calls use batch 1 and the same quantised base precision, with unmerged adapters. Concurrent CUDA compute is rejected. Repetitions are averaged within ID before paired group-bootstrap differences. Variant jobs are separate blocks, not a randomised crossover; thermal/background drift can remain. Serial requests/second is not service capacity.

4. Results

4.1. Model comparison

Table 3 combines the fixed-data LLMs, tuned encoders and sparse baselines. Qwen macro-F1 rises from 0.6295 to 0.8331 and Llama from 0.6247 to 0.8402. Seed SD is 0.0048/0.0020 for the adapted models. Their paired macro-F1 gains are 0.2035 [0.1934, 0.2141] and 0.2155 [0.2054, 0.2257], conditional on these fitted checkpoints.

Table 3. Internal test (15,428 rows). Macro-F1 brackets are 95% paired-group intervals; SD is across three actual optimisation refits. Base systems have no adaptation rows. TF-IDF SD zero denotes deterministic checks. Full-data TF-IDF has a larger training budget.

System	Train rows	Macro-F1 [CI]	Seed SD	Accuracy	Weighted F1
Qwen3 base	0	0.6295 [0.6182, 0.6405]	—	0.7352	0.7206
Qwen3 QLoRA	10,000	0.8331 [0.8245, 0.8411]	0.0048	0.8689	0.8678
Llama 3.1 base	0	0.6247 [0.6138, 0.6349]	—	0.7295	0.7177
Llama 3.1 QLoRA	10,000	0.8402 [0.8312, 0.8483]	0.0020	0.8736	0.8729
BERT	10,000	0.8119 [0.8028, 0.8203]	0.0010	0.8541	0.8528
DistilBERT	10,000	0.8104 [0.8014, 0.8188]	0.0065	0.8522	0.8511
RoBERTa	10,000	0.8105 [0.8015, 0.8187]	0.0039	0.8525	0.8520
TF-IDF/LR 10k	10,000	0.8068 [0.7973, 0.8157]	0.0000	0.8499	0.8474
TF-IDF/LR full	120,502	0.8476 [0.8402, 0.8549]	0.0000	0.8805	0.8798

At the matched 10k budget, QLoRA exceeds TF-IDF macro-F1 by 0.0263 and 0.0334, with paired intervals above zero (Table 4). Encoder differences from the sparse baseline are smaller, with macro-F1 intervals spanning zero. Full-data TF-IDF reaches 0.8476; Qwen's difference from it is -0.0145, whereas Llama's -0.0074 interval includes zero (unrounded upper bound 6.8×10^{-6}). This unequal-budget comparison rules out a general claim that QLoRA is the best classifier for the taxonomy. The consolidated points and intervals share one estimator; bootstrap means are not substituted for observed points.

Table 4. Internal macro-F1 paired differences, comparison minus reference. All intervals condition on fitted models. The final two rows are unequal-training-budget context.

Reference	Comparison	Difference	95% CI
Qwen3 base	Qwen3 QLoRA	0.2035	[0.1934, 0.2141]
Llama 3.1 base	Llama 3.1 QLoRA	0.2155	[0.2054, 0.2257]
TF-IDF/LR 10k	Qwen3 QLoRA	0.0263	[0.0199, 0.0327]
TF-IDF/LR 10k	Llama 3.1 QLoRA	0.0334	[0.0270, 0.0397]
TF-IDF/LR 10k	BERT	0.0051	[-0.0013, 0.0117]
TF-IDF/LR 10k	DistilBERT	0.0036	[-0.0025, 0.0098]
TF-IDF/LR 10k	RoBERTa	0.0037	[-0.0025, 0.0098]
TF-IDF/LR full	Qwen3 QLoRA	-0.0145	[-0.0221, -0.0071]
TF-IDF/LR full	Llama 3.1 QLoRA	-0.0074	[-0.0153, 0.0000]

4.2. Benchmark views, classes and output compliance

Unique-input-weighted benchmark macro-F1 is 0.6324/0.8210 for base/adapted Qwen and 0.6271/0.8272 for Llama. Paired gains are 0.1886 [0.1723, 0.2051] and 0.2000 [0.1838, 0.2166]. Matched-budget TF-IDF gives 0.7871; full-data TF-IDF gives 0.8347. The overlapping views support a dependence-aware source-related analysis, not twelve independent replications or external generalisation. Supplementary Sections S3–S4 give all metrics, seeds and per-view descriptions.

Figure 1 and the supplementary class tables describe the primary-comparison predictions. Debt-collection recall rises from 0.3262 to 0.7347 for Qwen and from 0.2754 to 0.7540 for Llama. Improvements are not uniform: Qwen credit-card recall changes from 0.7637 to 0.7550 despite its higher class F1. Of 15,428 internal predictions, Qwen base has one off-label output and each adapted refit has zero; Llama base and each refit have one. These are observed compliance counts, not enforced constrained decoding or evidence of prompt robustness. Residual confusions and different class supports remain important.

The dominant zero-shot error is the confusion of `debt_collection` with `credit_reporting`. Qwen makes this error 1,232 times before adaptation and 430.33 times after adaptation on average; the corresponding Llama counts are 1,167 and 403.33. These are reductions of 65.1% and 65.4%. Fine-tuning also reduces `retail_banking` → `credit_card` errors from 562 to 81 for Qwen and from 263 to 89 for Llama. Adapted counts are means of three refit count matrices; fractional values do not denote fractional complaints in a single run.

One residual issue remains visible in both models: `credit_reporting` → `debt_collection` errors increase from 142 to 344 for Qwen and from 87 to 336 for Llama. Higher debt-collection recall therefore coexists with more false assignments from the majority class. The main change is a redistribution of errors across neighbouring financial categories, rather than a uniform improvement in every decision boundary. These descriptive counts locate the remaining problem; they do not establish its causal mechanism or the operational cost of particular mistakes.

4.3. Same-checkpoint longer-input results

The 500/512 and 500/2,048 control predictions are identical. Increasing visible text to 1,500 characters improves adapted Qwen/Llama macro-F1 by 0.0072/0.0055 overall and by 0.0156/0.0117 in the longer-narrative cohort (Table 5). Zero-shot Qwen instead declines; the zero-shot Llama interval includes zero. The result supports a conditional benefit of longer input for these adapted checkpoints, not a universal improvement or an evaluation of longer-context training. Supplementary Section S5 also reports auxiliary length-stratum descriptions and matched 250/500-character benchmark views; their cohort and fitting conditions differ from this controlled inference comparison.

4.4. Controlled inference cost and supplementary sensitivity

For 500/512, mean end-to-end seconds are 0.2896/0.4120 for base/adapted Qwen and 0.2831/0.3633 for Llama (Table 6). The additional mean times are 0.1224 [0.1166, 0.1284] and 0.0802 [0.0758, 0.0847]. These differences include actual generated lengths and adapter execution, not an isolated adapter-overhead experiment. The balanced 100-record cohort and serial batch-one calls are not an operational workload benchmark. Full per-seed median/p95, token counts and all conditions are in Supplementary Section S6.

The separately declared Ref [14] conditions give internal macro-F1 0.8102 for BERT and 0.8095 for RoBERTa versus main values 0.8119/0.8105. Paired difference intervals span zero. The main RoBERTa condition uses 512 tokens. All source-specific parameter differences and three refits are reported in Supplementary

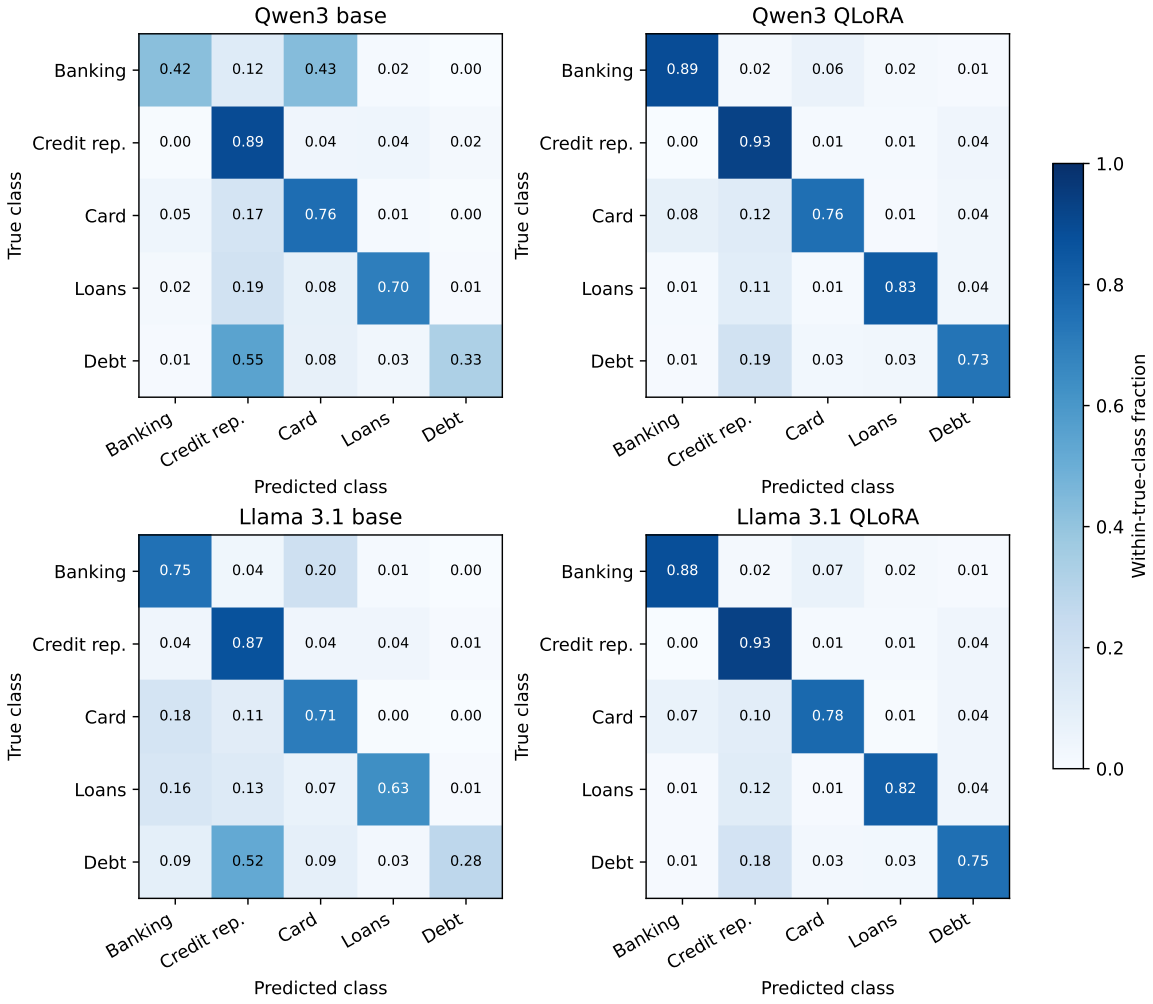


Figure 1. Internal confusion fractions, normalised within true class. QLoRA matrices average the three refit count matrices before normalisation; performance metrics are means of separately scored runs. Class order and off-label counts are supplied in Supplementary Section S3.

Table 5. Macro-F1 change for 1,500 versus 500 characters, both with a 2,048-token cap. "All" includes 15,428 rows; > 500 is the predefined 5,770-row cohort. QLoRA is the three-seed mean.

Model	Variant	Cohort	500	1,500	Difference	95% CI
Qwen3	base	All	0.6295	0.6224	-0.0072	[-0.0107, -0.0036]
Qwen3	QLoRA mean	All	0.8331	0.8403	0.0072	[0.0048, 0.0097]
Qwen3	base	> 500	0.6316	0.6135	-0.0181	[-0.0262, -0.0099]
Qwen3	QLoRA mean	> 500	0.8484	0.8640	0.0156	[0.0105, 0.0209]
Llama 3.1	base	All	0.6247	0.6227	-0.0020	[-0.0058, 0.0018]
Llama 3.1	QLoRA mean	All	0.8402	0.8456	0.0055	[0.0029, 0.0081]
Llama 3.1	base	> 500	0.6432	0.6415	-0.0017	[-0.0102, 0.0068]
Llama 3.1	QLoRA mean	> 500	0.8562	0.8679	0.0117	[0.0065, 0.0172]

Table 6. Fresh-call end-to-end latency (seconds), 500 characters/512 tokens, RTX 5060 Ti. QLoRA mean averages the three variant means; paired intervals use within-ID repetition means.

Model	Base mean	QLoRA mean	Added time	95% CI
Qwen3	0.2896	0.4120	0.1224	[0.1166, 0.1284]
Llama 3.1	0.2831	0.3633	0.0802	[0.0758, 0.0847]

Section S7 without selecting a replacement on test performance.

5. Discussion

Under a fixed 10k adaptation budget, the two LLM families show substantial within-model gains and smaller positive margins over the matched sparse classifier. Full-data TF-IDF remains at least competitive in observed macro-F1, with substantially more labelled training data. The practical question is therefore conditional on labelled-data availability, the value of retaining a generative model, and measured inference cost. These experiments do not test cross-task reuse or a changed taxonomy, so such potential workflow benefits are motivations rather than findings.

Model-visible separation is central to the comparison because full-text duplicate detection alone does not identify all collisions under truncation. Grouping the visible representations and excluding fitting conflicts provides an explicit basis for the paired evaluation. The auxiliary dependence analyses in Supplementary Sections S5 and S8 quantify the limitations of full-text grouping; their different memberships, model identities and environments preclude attributing cross-condition score differences solely to overlap handling.

The class-level pattern is particularly relevant for complaint triage. Debt collection narratives may mention disputed accounts, credit bureaus and credit reports, so the five-label task has closely related categories. The reduction in debt-to-reporting errors is accompanied by increased reporting-to-debt errors, which helps explain why a gain in aggregate macro-F1 does not resolve every routing concern. The balanced benchmark views expose minority-class behaviour, while the imbalanced views retain different class prevalences; their overlap requires the shared-group analysis rather than treating them as independent confirmations. Human review, cost-sensitive training and class-specific calibration are possible directions for studying these residual errors, not evaluated safeguards in the present experiments.

Related work already establishes complaint LLM fine-tuning. Our distinction from Güllü et al. is the explicit NF4/LoRA and completion-only protocol, paired same-model baseline, fixed adaptation membership and separated uncertainty; from Roumeliotis et al., it is the adaptation comparison beyond zero-shot evaluation. Different splits, metrics and budgets preclude cross-paper performance ranking. Longer input helps the adapted checkpoints here, while zero-shot results show that more text alone is not a general solution. Latency intervals quantify conditional measured costs; their operational acceptability requires workload-specific evidence.

6. Limitations

Evaluation is retrospective on previously inspected CFPB-related records; there is no independent-source, temporal or prospective generalisation test. Exact-input audits and conservative groups do not eliminate semantic duplication. Bootstrap intervals condition on fixed refits and the chosen protocol, excluding training/model-selection uncertainty. Three seeds give limited evidence about training variation, and multiple intervals are not family-wise adjusted. The 10k budget, fixed prompt and two 8B model families do not establish an optimal sample size, prompt robustness or universal classifier ranking.

The longer-input comparison changes inference text only; the models were trained with at most 500 characters. Latency uses serial unmerged-adapter jobs, a balanced 100-record sample and separate variant

blocks, leaving thermal/background drift and output-length effects. Encoder/TF-IDF service latency and energy or total training cost were not benchmarked here. Upstream CFPB-to-Kaggle preprocessing is unavailable. Some auxiliary fits also lack complete fitted weights and immutable upstream identities, limiting their reconstruction; Supplementary Section S8 states this scope separately from the fully identified primary models.

7. Conclusion

This study quantifies a reproducible same-model transition from zero-shot prompting to QLoRA under a fixed 10k training budget. Internal macro-F1 gains of 0.2035/0.2155 coexist with competitive sparse baselines, dependence among source-related benchmark views and measurable inference costs. The matched-data comparisons, full-data context, seed variation and longer-input results support a conditional adaptation trade-off. The value of the study is the measured within-model transition from zero-shot inference to competitive fixed-budget supervised performance, together with a transparent account of data dependence, class errors and inference cost. This provides an empirical basis for comparing an adapted generative model with lighter classifiers when designing a complaint-triage workflow. They do not establish first use of fine-tuned LLMs for financial complaints, general classification superiority or deployment/regulatory readiness.

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Declaration of Generative AI and AI-assisted technologies, if any

AI-assisted tools supported analysis-code preparation, literature comparison, wording and LaTeX formatting. Numerical results come from recorded computations and independently checked prediction files. The authors retain responsibility for source verification, interpretation, final review and the submitted manuscript.

Data availability statement

The CFPB complaint database is publicly accessible. Source/derived data, fixed selections, code, configurations, model identities, validation and final predictions, and statistical evidence are provided in the reproducibility repository at `https://github.com/muhammettan28/CFPB_tuning`. Supplementary Section S8 describes its contents and the CPU result-verification procedure. The repository excludes third-party literature and neural-model weights. Limitations concerning upstream preprocessing and auxiliary-model reconstruction are documented in Supplementary Section S8.

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